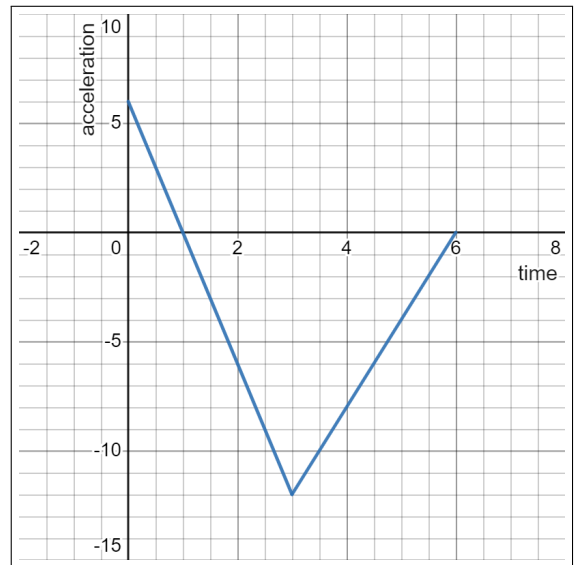


1. (Spring 2023, #9) You're driving a car at a speed of 30 ft/sec, and notice a cone ahead in the road. The graph to the right gives your acceleration  $t$  seconds after you see the cone (measured in ft/sec<sup>2</sup>).



- (a) What is your velocity at  $t = 3$  seconds?

- (b) Find an expression for the velocity,  $v(t)$ , from  $t = 3$  to  $t = 6$ .

- (c) How far did the car travel from  $t = 0$  to  $t = 6$ ?

2. (Fall 2021, #1) Let  $a(t)$  denote acceleration of a particle,  $v(t)$  denote its velocity, and  $s(t)$  denote its position. Suppose that  $a(t) = 2t + 10e^{5t}$ .

(a) If  $v(0) = 8$ , find  $v(t)$ .

(b) If  $s(1) = 3$ , find  $s(t)$ .

3. (Fall 2016, #7) Assume that the acceleration (in meters per second squared) of a particle at time  $t$  (in seconds) is given by

$$a(t) = \sin\left(\frac{t}{2}\right).$$

The particle is at position  $x = 1$  meters at time  $t = 0$  seconds and it is at position  $x = 8$  meters at time  $t = \pi$  seconds. Find the position  $x(t)$  of the particle as a function of time. (Recall that the acceleration of a particle is the derivative of its velocity.)

4. (Fall 2011, # 10) Suppose that the acceleration of a particle (in m/sec<sup>2</sup>) at time  $t$  (in sec) is given by the function

$$a(t) = 6t - 6$$

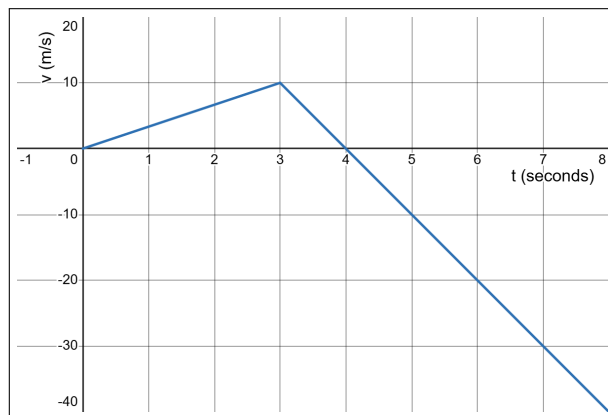
for  $t \geq 0$ .

- (a) Find the velocity of  $v(t)$  of the particle as a function of time if  $v(0) = -9$  m/sec.

- (b) Find the net displacement of the particle from  $t = 0$  to  $t = 4$ , i.e., the difference in the particle's position between  $t = 0$  and  $t = 4$ . Include units in your answer.

- (c) Find the total distance traveled by the particle from  $t = 0$  to  $t = 4$ . Include units in your answer.

5. (Fall 2002, #8) Shown below is the graph of the velocity of a toy rocket that is launched into the air, uses up all its fuel, and falls back to the ground.



(a) When does the rocket reach its highest point in the air?

(b) How high is it at that point in time?